

Course Framework

Program Name	Full Stack Web Developer
Course Name	.NET
Name of Educational Developer	Angela Tessier
Name of SME(s)	Naresh Nagandla
Course Hours	60 hrs
Course Delivery	Online Instructor-Guided
Course Description	This course builds on the C# programming course, providing you with lessons on how to develop cross-platform applications using .NET. We explore new features such as working with Files, Streams and Serialization, and working with data using Entity Framework core. Additionally, we explore how data is secured using encryption and hashing. Finally, we will learn how to use LINQ syntax for querying data collections.
Course Rationale	In today's competitive technological landscape, the ability to develop scalable, efficient, and modern applications is a valuable skill set. This course has been constructed to provide participants with understanding of the .NET framework, empowering them to build diverse applications ranging from desktop and web applications to cloud-based services.
Course Outcomes and Sub Outcomes	Learning Sub-Outcomes: Outcome 1: Utilize fundamental .NET data types: numbers, text, collections, arrays to construct a function. - Describe how C# keywords are related to .NET types, and the relationship between namespaces and assemblies - Describe the difference between primitive and non-primitive data types

	Outcome 2: Demonstrate how to Publish .NET apps and libraries for use cross-platform. • Describe how NuGet packages work • Demonstrate how to publish applications for deployment • Demonstrate how to package libraries for NuGet distribution • Describe how to internationalize a .NET application Outcome 3: Demonstrate how to Read and write to files and streams, text encoding and serialization. • Demonstrate how to manage the file system programmatically • Demonstrate how to read and write with streams • Demonstrate how to serialize and deserialize objects • Demonstrate how to encrypt and decrypt data • Describe how to Hash data • Describe how to generate random numbers • Describe how to Authenticate and authorize users • Describe how to protect data from being viewed by malicious users using encryption, and from being manipulated or corrupted using hashing and signing Outcome 4 : Read and write to databases, such as Microsoft SQL Server, SQLite, and Azure Cosmos DB, using the object-relational mapping technology named Entity Framework Core (EF Core). • Demonstrate how to set up the Entity Framework Core • Define Entity Framework Core models • Demonstrate how to query an Entity Framework Core model • Demonstrate how to manipulate data with EF Core Outcome 5: Utilize Language Integrated Query (LINQ), a set of language extensions that add the ability to work with sequences of items and then filter, sort, and project them into different outputs. • Describe how to use LINQ with Entity Framework Core • Create a LINQ extension methods
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	Demonstrate how to query sub-sets of data collection using LINQ
Overall Assessment Scheme	Assignments- 100%
Assessment Breakdown	Assignment 1 : Pattern Matching with Regular Expressions 25% of overall Grade Assignment 2 : File Streams and Serialization in .NET 25% of overall Grade Assignment 3 : Querying with LINQ (Language Integrated Query) 25% of overall Grade Quiz 1 : File Operations, Stream Handling and Serialization Concepts in the context of .NET. - 25% of overall Grade
Required Textbook	Price, M. J. (2024). <i>C# 11 and .NET 7 – Modern Cross-Platform Development Fundamentals (7th ed.)</i> . Packt Publishing
Required Courseware (from Publisher)	N/A
Required Software (Not from Publisher)	N/A
Required Equipment	N/A

Assessment Scheme

Assessment	Description	Weight
Assignment 1: Pattern Matching with Regular expressions	Students will create a program that checks the patterns in Regular Expression	25%
Assignment 2: File Streams and Serialization in .NET	Students will create a program that implements serialization to save it to the file system as XML.	25%
Assignment 3: Querying with LINQ (Language Integrated Query)	Students will implement program that queries the database using Entity Framework Core and displays the data using LINQ (Language Integrated Query)	25%
Quiz 1: File Operations, Stream Handling and Serialization Concepts in the context of .NET.	Students will be assessed on their understanding of file operations, stream handling, and serialization concepts in the context of .NET. It covers a range of topics from basic file operations to advanced serialization techniques.	25%

Assessment Mapping

Assessment Type	Outcome 1: Utilize fundamental .NET data types: numbers, text, collections, arrays to construct a function	Outcome 2: Demonstrate how to Publish .NET Core apps and libraries for use cross-platform	Outcome 3: Demonstrate how to Read and write to files and streams, text encoding and serialization	Outcome 4: Read and write to databases, such as Microsoft SQL Server, SQLite, and Azure Cosmos DB, using the object-relational mapping technology named Entity Framework Core (EF Core)	Outcome 5: Utilize Language Integrated Query (LINQ), a set of language extensions that add the ability to work with sequences of items and then filter, sort, and project them into different outputs
Assignment 1: Pattern Matching with Regular expressions	x				
Assignment 2: File Streams and Serialization in .NET		x	x		
Assignment 3: Querying with LINQ (Language Integrated Query)				x	x
Quiz 1: File Operations, Stream Handling and Serialization Concepts in the context of .NET.	x	x	x	x	x

Module/ Topic	Course Learning Outcome and Sub-outcomes	Summa tive Assess ments (includ e title, descrip tion and grade points/ weighti ngs)	Activities (Facilitated, Peer-to-Peer & Independent)	Textbook Chapters, Resources and Materials
Module 1 .NET Basics 20 Hours In this module, we will explore the features within the .NET framework. .NET developers have access to a diverse range of functionalities that extend well beyond basic language concepts. Our focus will be	Outcome : Utilize fundamental .NET data types: numbers, text, collections, arrays to construct a function. Sub-Outcomes: - Describe how C# keywords are related to .NET types, and the relationship between namespaces and assemblies - Describe the difference	Assignment1.docx Weight: 25%	This Module lasts one week and will require 20 hours of time to complete this module. In that time, you are required to: - Read Chapter 7 and Chapter 8 from the course textbook - Watch the supplementary videos in the Module Readings section - Introduce yourself to your classmates in the discussion forum.	Module1 Content: .NET Basics.pptx External Videos : Title: What is .NET Youtube Link: What is .NET Watch the video What is .NET. This video will discuss the following about .NET such as what it is, what languages and tools you can use with it, what you can build with it, and more! Title: Visual Studio Code vs Visual Studio

understanding the standard types available in .NET, along with the process of utilizing .NET for application development and cross-platform deployment. We will also study the efficient sharing and reuse of code through .NET Standard class libraries and NuGet. By harnessing the capabilities of NuGet packages, developers open up limitless possibilities for reusing code that has been previously implemented and rigorously tested.	between primitive and non-primitive data types Outcome : Demonstrate how to Publish .NET apps and libraries for use cross-platform. Sub-Outcomes: • Describe how NuGet packages work • Demonstrate how to publish applications for deployment • Demonstrate how to package libraries for NuGet distribution • Describe how to internationalize a .NET application		• Complete the practice exercises, post to the discussion forum, and provide feedback to your classmates • Complete Assignment #1 If you have any difficulties with the material in this chapter, please post to the Module 1 - Help Topic	Youtube Link: Visual Studio Code vs Visual Studio Watch the video Visual Studio Code vs Visual Studio. Deciding on Visual Studio or Visual Studio Code is not just a personal decision but based on the best option for the operating system and application type you are building. Take note of the other relevant points from the video, and list any questions you have. Post a minimum of one question in the Module 1 Help forum for your classmates/instructor to respond to. Title : What's New in C# 12 Youtube Link : What's New in C# 12 .NET Conf 2023 Watch the video “What’s new in C# 12” which will take you through the new features in C# 12. Take note of the other relevant points from the video, and list any questions you have. Post a minimum of one question in the Module 1 Help forum for your classmates/instructor to respond to. Title : Introduction to .NET Package Management using NuGet Youtube Link : Introduction to .NET Package Management ...
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				Watch the video Introduction to .NET Package Management using NuGet. Watch this video that discusses open source package management using NuGet, the official package manager tool for .NET. Take note of the other relevant points from the video, and list any questions you have. Post a minimum of one question in the Module 1 Help forum for your classmates/instructor to respond to. Activities: Discussion [Non-Graded] : NuGet Package Management This discussion forum is designed for you to share your thoughts, experiences, and questions related to packaging and distributing .NET applications. Feel free to engage with the following topics or start new discussions based on your interests and experiences. • How has NuGet simplified the management of dependencies in your .NET projects? • Share examples of NuGet packages that have been instrumental in your development work. • Discuss any tips or challenges you've faced when using NuGet for package management.
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				Discussion [Non-Graded] : Building and Packaging .NET Applications Share your insights into best practices, tools, and considerations for different platforms. • What are the key steps you follow when building and packaging a .NET application for distribution? • Share experiences with cross-platform deployment and any compatibility considerations. • Discuss tools or techniques that have helped automate the build and packaging process. Practice Activity : W Module1 - Practice Activity.docx Knowledge Check (Non Graded) : W Module1 - Knowledge Check(Non-Graded).d...
Module2 Files, Streams and Serialization 20 Hours In this module, we will learn some advanced concepts, learning how to get	Outcome : Demonstrate how to Read and write to files and streams, text encoding and serialization. Sub-Outcomes :	Assignment 2: Module2 - Assignment.docx	This Module lasts one week and will require 20 hours of time to complete this module. In that time, you are required to: • Read Chapter 9 from the course textbook	Module2 Content: Files Data and Security.pptx External Links : Title: File and Stream I/O External URL: https://learn.microsoft.com/en-us/dotnet/standard/io/

data, how to move data, and how to keep it safe from malicious users. We will also learn how to work with the filesystem to read and write files and also how to work with data streams. We will examine the basic concepts of security and cryptography and how they can be practically applied in .NET. It's important to always write code with security in mind and knowing how data is protected will help you make smart decisions. Finally, we will examine user management. This module will explore the difference between authorization and authentication and what this means when developing applications because you will need an	• Demonstrate how to manage the file system programmatically • Demonstrate how to read and write with streams • Demonstrate how to serialize and deserialize objects • Demonstrate how to encrypt and decrypt data • Describe how to Hash data • Describe how to generate random numbers • Describe how to Authenticate and authorize users • Describe how to protect data from being viewed by malicious users using encryption, and from being manipulated or	Weight: 25%	• Watch the supplementary videos in the Module Readings section • Complete the practice exercises, post to the discussion forum, and provide feedback to your classmates • Complete Assignment #2 If you have any difficulties with the material in this chapter, please post to the Module 2 - Help Topic	Please go through the above link which covers the topics related to file and stream operations, providing information, guidance and examples for developers working with I/O in .NET. External Links : Title: Serialization in .NET External URL: https://learn.microsoft.com/en-us/dotnet/standard/serialization/ Please go through the above link which covers the concepts of serialization, how to serialize and deserialize objects, handling different formats (such as XML and JSON) and considerations for versioning and backward compatibility in serialization. External Link : Title Welcome to System.Text.Json URL : Welcome to System.Text.Json - How to read ... Please go through the above link which walks through the basics of System.Text.Json to read and write .NET objects as JSON External Links :
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understanding of what's needed to support a multi-user application securely.	corrupted using hashing and signing			Title: Concepts of cryptography External URL: Describe-concepts-of-cryptography Please go through the above link which covers fundamental cryptographic concepts, encryption and decryption techniques, key management, and other relevant topics related to securing data through cryptographic methods. Activities: Discussion [Non-Graded] : File Operations in .NET Share your experiences and challenges when working with files. Discuss common file operations such as creation, reading, writing, and deletion. How do you handle file paths, and what strategies do you use to ensure robust file management in your projects? • What are the fundamental file operations you frequently perform in .NET projects? • Share any tips for managing file paths and handling exceptions during file operations. • Discuss scenarios where asynchronous file operations might be advantageous.
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				Discussion [Non-Graded] : Serialization Strategies in .NET Serialization is essential for converting object data into a format suitable for storage or transmission. Let's discuss various serialization techniques in .NET, such as binary, XML, and JSON serialization. Share your preferences and experiences with different serialization formats. • In what scenarios would you choose binary serialization over XML or JSON, and vice versa? • Discuss the advantages and challenges of XML serialization in .NET. • How do you handle versioning and backward compatibility when dealing with serialized data? Practice Activity : W Module2 - Practice Activity.docx Knowledge Check (Non Graded) : W Module2 - Knowledge Check (Non-Graded)....
Module 3 Databases, EFCore and LINQ 20 Hours	Outcome : Read and write to databases, such as Microsoft SQL Server, SQLite, and Azure Cosmos DB, using the object-relational	Assignment 3: Modul e3 - Assign	This Module lasts one week and will require 20 hours of time to complete this	Module 3 Content P Databases EF Core and LINQ.pptx External Videos :

In this module, we will study the processes through which .NET manages and retrieves data, employing a tool known as Entity Framework Core (EF Core). To integrate data from databases into your application, it's essential to establish a connection, formulate queries to fetch the required data, and establish mappings between database tables and objects for use in code. EF Core handles these tasks, freeing you to concentrate on essential business logic and application-specific intricacies. Then we will study LINQ (Language Integrated Query). LINQ allows you to retrieve data from the database as needed and	mapping technology named Entity Framework Core (EF Core). Sub-Outcomes : • Demonstrate how to set up the Entity Framework Core • Define Entity Framework Core models • Demonstrate how to query an Entity Framework Core model • Demonstrate how to manipulate data with EF Core Outcome : Utilize Language Integrated Query (LINQ), a set of language extensions that add the ability to work with sequences of items and then filter, sort, and project them into different outputs.	ment.d ocx Weight: 25% Quiz : W Qu... Weight: 25%	module. In that time, you are required to: • Read Chapter 10 and 11 from the course textbook. • Watch the supplementary videos in the Module Readings section. • Complete the practice exercises and post to the discussion forum, and provide feedback to your classmates. • Complete Assignment #3 and Assignment #4. If you have any difficulties with the material in this chapter, please post to the Module 3 - Help Topic	Title: Getting started with Entity Framework Core Youtube Link: Entity Framework Core for Beginners Watch the video Getting Started with EF Core. Summarize the relevant topics from the video, and list any questions you have. Post a minimum of one question, and participate in the discussions with your instructor and classmates through the Module 3 - Help Forum. Title: Working with Existing Databases Youtube Link: Entity Framework Core for Beginners Watch the video Working with an Existing Database. The video shows how to reverse engineer an existing database for use with EF Core. Summarize the other relevant points from the video, and list any questions you have. Post a minimum of one question, and participate in the discussions with your instructor and classmates through the Module 3 - Help Forum. Title: Introduction to LINQ Youtube Link: Introduction to Language Integrated Query
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manipulate data within your application using principles reminiscent of SQL (Structured Query Language). Lastly, we'll examine the straightforward process of serializing data into XML (Extensible Markup Language) using libraries accessible in .NET.	Sub-Outcomes: • Describe how to use LINQ with Entity Framework Core • Create a LINQ extension methods • Demonstrate how to query sub-sets of data collection using LINQ			Watch the video Introduction to LINQ. The video describes what LINQ is, where you can explore LINQ syntax on your own, and how LINQ makes it easy to query any data source. Summarize the other relevant points from the video, and list any questions you have. Post a minimum of one question, and participate in the discussions with your instructor and classmates through the Module 3 - Help Forum. Title: LINQ Syntax Youtube Link: LINQ Query Syntax Watch the LINQ Syntax video. The video explores the syntax used in LINQ queries, and the palette of tools you can use to query data to find, filter, or analyze data sources. Summarize the other relevant points from the video, and write any questions you have. Post a minimum of one question in the Module 3 - Help Forum and participate in the discussion with your instructor and classmates. Title: LINQ Query Composition and Execution Youtube Link: LINQ Query Composition and Execution Watch the video LINQ Query Composition and Execution. The video shows how LINQ composes queries written in multiple statements, or even
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				multiple methods, into a single query operation. Summarize the other relevant points from the video, and list any questions you have. Post a minimum of one question, and participate in the discussions with your instructor and classmates through the Module 3 - Help Forum. External Links : Title: Retrieve relational data by using Entity Framework Core External URL: persist-data-ef-core Please go through the above link which covers fundamental concepts of Entity framework code using an Object Relational Mapper(ORM). Activities : Discussion [Non-Graded] : Entity Framework Core Essentials Discuss the fundamentals of Entity Framework Core. Share your experiences with database connectivity, model creation, and how EF Core streamlines database operations in .NET applications. • What challenges have you faced when establishing database connections in EF Core? • Share tips for efficient model creation and management in Entity Framework Core.
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				- Discuss real-world scenarios where Entity Framework Core proved to be a powerful tool. Discussion [Non-Graded] : LINQ Fundamentals Explore the foundational concepts of Language-Integrated Query (LINQ) in .NET. Share your understanding of LINQ basics, its syntax, and discuss scenarios where LINQ can be advantageous in simplifying data manipulation. - What is your understanding of the fundamental concepts of LINQ? - How does LINQ enhance code readability and maintainability in data manipulation tasks? - Share examples of LINQ queries that showcase its syntax and simplicity. Practice Activity : W Module3 - Practice Activity.docx Knowledge Check (Non Graded): W Module3 - Knowledge Check (Non-Graded)....
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